

Generative AI Makes Creative Output More Homogeneous

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Does Generative AI Make Us Think Alike? A Systematic Review and Meta-Analysis of Homogenization Effects in Human–AI Co-Creation

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Does generative AI make us think alike? As human–AI co-creation becomes embedded in creative practice, concerns have emerged that creative outputs may exhibit homogenization, as reduced diversity of meanings, across users, even when individual creativity improves. We address this concern through a systematic review and three-level meta-analysis of empirical studies on homogenization in human–AI co-creation (19 studies, 61 effect sizes). Results reveal a small but statistically significant homogenization effect associated with AI use, robust across sensitivity analyses and not explained by publication bias. Moderator analyses indicate that homogenization is task-sensitive, with stronger effects in semantically constrained ideation tasks than in minimally constrained divergent thinking tasks. Additional analyses suggest that homogenization may persist beyond co-creative episodes and extend to real-world contexts. Rather than indicating a loss of creativity, these findings point to a gradual, scale-dependent reorganization of creative diversity shaped by task structure, interaction workflows, and sociotechnical feedback processes.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Artificial intelligence**.

Additional Key Words and Phrases: Generative Artificial Intelligence, Homogenization, Human–AI Co-Creation, Meta-Analysis

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1 Introduction

The rapid integration of generative artificial intelligence (GenAI) into creative practice has intensified interest in how creativity is shaped by computational support. Large language models (LLMs) and related generative systems are now routinely embedded in tools for ideation, writing, design, and artistic production [8, 54]. Within Human–Computer Interaction (HCI) and creativity support research, early work primarily framed such technologies in terms of *augmentation*, emphasizing gains in fluency, efficiency, and accessibility [28, 55]. GenAI systems were often positioned as resources to accelerate ideation, broaden participation, or help overcome familiar creative impasses such as fixation, long documented in human ideation [27]. For several years, HCI research thus largely approached GenAI as a general interaction technology for creativity support.

Recently, empirical work has raised a more ambivalent concern surrounding the widespread adoption of GenAI in creative practice, particularly in *human–AI co-creation*. By ‘co-creation,’ we refer to situations in which humans and generative systems jointly contribute to the emergence, elaboration, and stabilization of creative artifacts, such that

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outcomes cannot be meaningfully attributed to either human intention or algorithmic generation alone [13, 28]. The key concern motivating this paper is whether such human–AI co-creation leads to *homogenization*. By ‘homogenization,’ we mean a *reduced diversity of meanings* in creative output at the *collective* level, rather than a decline in individual creativity [4, 19, 45]. Addressing this perceived concern about homogenization reflects a broader shift away from asking whether GenAI helps or hinders creativity in general, and toward examining how co-creative engagement with GenAI may *reshape* the distribution of creative outcomes across groups and time [30].

Motivated by this concern, this paper asks the following *main research question (RQ)*: *does human–AI co-creation lead to homogenization at the collective level?* To address this, the paper synthesizes the emerging empirical literature at the intersection of HCI and creativity research through a *systematic review* and *meta-analysis* of the first wave of studies (2022–2026). The paper begins by developing the theoretical and empirical background, concluding with a refinement of the main research question into three sub-questions concerning task sensitivity, temporal persistence, and ecological validity. It then presents the systematic review and meta-analytic methods, followed by results and a discussion of implications, particularly for HCI research.

2 Background

Concerns about whether co-creative engagement with generative systems may lead to homogenization, understood as *convergence* in the *semantic* or *conceptual space* of creative outputs, are not entirely new. Long before the advent of GenAI, creativity research documented how exposure to examples, precedents, or dominant solution patterns can constrain ideation through design fixation or example-based interference [15, 27, 57]. What appears novel in the context of GenAI is therefore less the underlying cognitive mechanism, but rather the *scale* and *synchronicity* of exposure: many humans now co-create, often repeatedly, with the same or closely related GenAI systems. This raises the question of whether such processes may affect creative cognition at the collective level.

2.1 Individual Gains versus Collective Consequences

A key issue concerns the relationship between individual-level benefits and collective-level consequences. Controlled experiments consistently suggest that people engaged in human–AI co-creation produce artifacts that are rated as more creative, higher quality, or more enjoyable on average, particularly among less-experienced creators [19, 48]. At the same time, these artifacts tend to become more similar to one another across users [39, 45]. Such findings challenge an often implicit assumption within creativity support research: that improvements at the individual level naturally scale to system-level benefits. Creativity and innovation research, however, has long suggested otherwise. In studies of brainstorming and ideation, the value of group ideation often resides less in mean idea quality than in variance, recombination potential, and the emergence of highly distinctive ideas [23, 52]. From this perspective, homogenization constitutes a distinct *risk* of human–AI co-creation, even when average creative performance appears to improve. Similar concerns were voiced in earlier debates on computational creativity, where increasing generative capacity raised questions about stylistic convergence and cultural flattening [10, 12].

2.2 Creativity as a Systemic and Sociocultural Phenomenon

Creativity research provides a well-established theoretical basis for understanding why homogenization at the collective level matters. While creativity is often defined as a conjunction of originality and usefulness [51], scholars since at least the 1990s have emphasized that creativity is a *systemic*, *sociocultural*, and *distributed* phenomenon [2, 16]. In systems-oriented accounts, creativity emerges through interactions among individuals, symbolic domains, and social

fields that evaluate and legitimate novelty [16]. Sociocultural frameworks such as the Five As [24], along with integrative perspectives, further highlight collaboration, circulation, and uptake of creative products as constitutive rather than peripheral aspects of creativity [41, 60]. Across these perspectives, creativity depends not only on generating novel artifacts, but on sustaining plurality, disagreement, and adaptive variation over time [25]. Homogenization therefore constitutes a potential *system-level failure mode*: a creative ecology may continue to produce outputs that appear novel in isolation while the underlying opportunity space for exploration and recombination gradually narrows [10, 31].

2.3 Cognitive Mechanisms: Fixation, Memory, and Convergence

Since the seminal work of Finke, Ward, and Smith [20], cognitive creativity research has offered explanations for how convergence, as reduced diversity of meanings, can arise over time. Recent integrative frameworks describe creative ideation as a dynamic balance between exploration and exploitation, mediated by semantic memory, episodic recombination, and cognitive control [7, 9]. Human-AI co-creation intervenes directly in these mechanisms. By producing fluent, elaborated, and semantically coherent suggestions, GenAI acts as a strong semantic attractor that reduces uncertainty during ideation [9]. When many humans co-create with similar LLMs—and exchange and apply best-practice prompting strategies—these attractors become shared points of reference, amplifying *collective* reduced diversity of meanings beyond individual fixation [4, 64].

2.4 Human-AI Co-Creation as a Sociotechnical Feedback Loop

From an HCI perspective, homogenization increasingly appears as an *emergent property* of human-AI co-creative systems. Across ideation, writing, and design tasks, engagement in human-AI co-creation has been shown to increase fluency or perceived creativity while reducing between-participant variance [19, 45]. Research on collective ideation systems further demonstrates that algorithmic suggestions can unintentionally drive convergence unless the diversity of meanings is explicitly scaffolded or requested [14, 56]. These effects align with intrinsic properties of contemporary pre-trained GenAI, such as mode collapse and preference-driven convergence [8, 58]. When embedded in creative workflows, this produces a *sociotechnical feedback loop* in which models encode historical norms, humans adapt to model outputs, and repeated co-creation stabilizes shared semantic trajectories [66].

2.5 Task Sensitivity, Temporality, and Ecological Validity

Homogenization effects are unlikely to be uniform across creative contexts. Contemporary creativity research suggests that tasks with richer semantic framing and stronger evaluative constraints, such as ideation, writing, or design, are more susceptible to convergence, as reduced diversity of meanings, than minimally constrained divergent thinking tasks [7, 31]. Recent HCI studies provide converging, though still limited, evidence that homogenization effects during human-AI co-creation are more pronounced in such semantically rich tasks than in open-ended ones [1, 4]. While individual studies examine homogenization within specific tasks, a *systematic comparison* of homogenization effects across creative task types has, to our knowledge, not yet been undertaken.

Creativity is also increasingly understood as *trajectory-based* rather than episodic. Early convergence during human-AI co-creation may limit subsequent exploration, and longer-term engagement with generative systems may reshape how creative problems are framed and how semantic memory is accessed [9, 50]. In studies assessing *post-AI-use effects*, homogenization is not measured merely as a continuation of the same creative process, but often in subsequent instantiations of similar tasks. This suggests that what may carry over is a *narrowing* in creative thinking or semantic

framing, rather than persistence within a single ideation episode. This distinction is critical, i.e., framing post-AI-use homogenization as a generalization effect *across* creative situations rather than a short-lived artifact of task repetition.

Although most current findings stem from laboratory experiments, quasi-experimental studies of real-world corpora suggest that homogenization effects may extend beyond controlled settings [46, 69, 70]. Taken together, these findings underscore the importance of assessing the *ecological validity* of homogenization effects in human-AI co-creation.

2.6 Research Questions

This study contributes a *systematic review* and *meta-analysis* of this emerging empirical literature on homogenization effects in human-AI co-creation. Informed by current HCI and creativity research, this study does not presume homogenization as either an inevitable consequence of GenAI usage or a negligible lab artifact. Instead, it synthesizes available empirical evidence across creative task types, temporal framings, and methodological settings to better understand this development. We examine *three sub-research questions (SQs)* to *characterize* this homogenization effect:

- **SQ1.** Do homogenization effects in human-AI co-creation differ *across creative task types* (e.g., divergent thinking, ideation, writing, visual design)?
- **SQ2.** Do homogenization effects *persist* after episodes of human-AI co-creation end, indicating effects that extend beyond immediate co-creative engagement?
- **SQ3.** Are homogenization effects primarily a lab phenomenon, or do comparable patterns manifest in *real-world* creative contexts?

3 Method

To examine these research questions, a systematic review and meta-analysis were conducted based on current best practices.

3.1 Search

A systematic search was conducted across multiple databases to identify articles eligible for inclusion in the meta-analysis. We searched Web of Science, ACM Digital Library, and additionally Google Scholar (first 200 results, sorted by relevance) and the preprint and dissertation databases arXiv, PsyArXiv, SSRN, and ProQuest Dissertations to capture gray literature. Databases were searched using a predefined Boolean query combining search terms related to GenAI, creative output, homogenization effects, and experimental methodology. Where available, titles and abstracts were queried for records published between 2022, the year of ChatGPT's public release and widespread adoption of pre-trained GenAI, and February 25, 2026, as the cut-off date, covering the initial wave of empirical studies on homogenization effects in human-AI co-creation [18]. In total, 629 records were retrieved (ACM Digital Library = 58; PsycINFO = 19; Web of Science = 273; Google Scholar = 200; arXiv = 71; PsyArXiv = 4; SSRN = 0; ProQuest Dissertations = 4). Through de-duplication in Zotero (v7.0.9), 36 records were removed, leaving 593 records for screening.

3.2 Eligibility Criteria

Articles reporting experimental studies comparing participants using GenAI to those not using it, for homogenization effects at the collective level, were eligible. GenAI could include pre-trained LLMs, image, music, or audio generators. AI uses broadly entailed human-AI co-creation, including generating guidance, feedback, or ideas. Studies testing no clear human-AI co-creative interaction were excluded, e.g., [33, 58, 66]. Inclusion further required an outcome

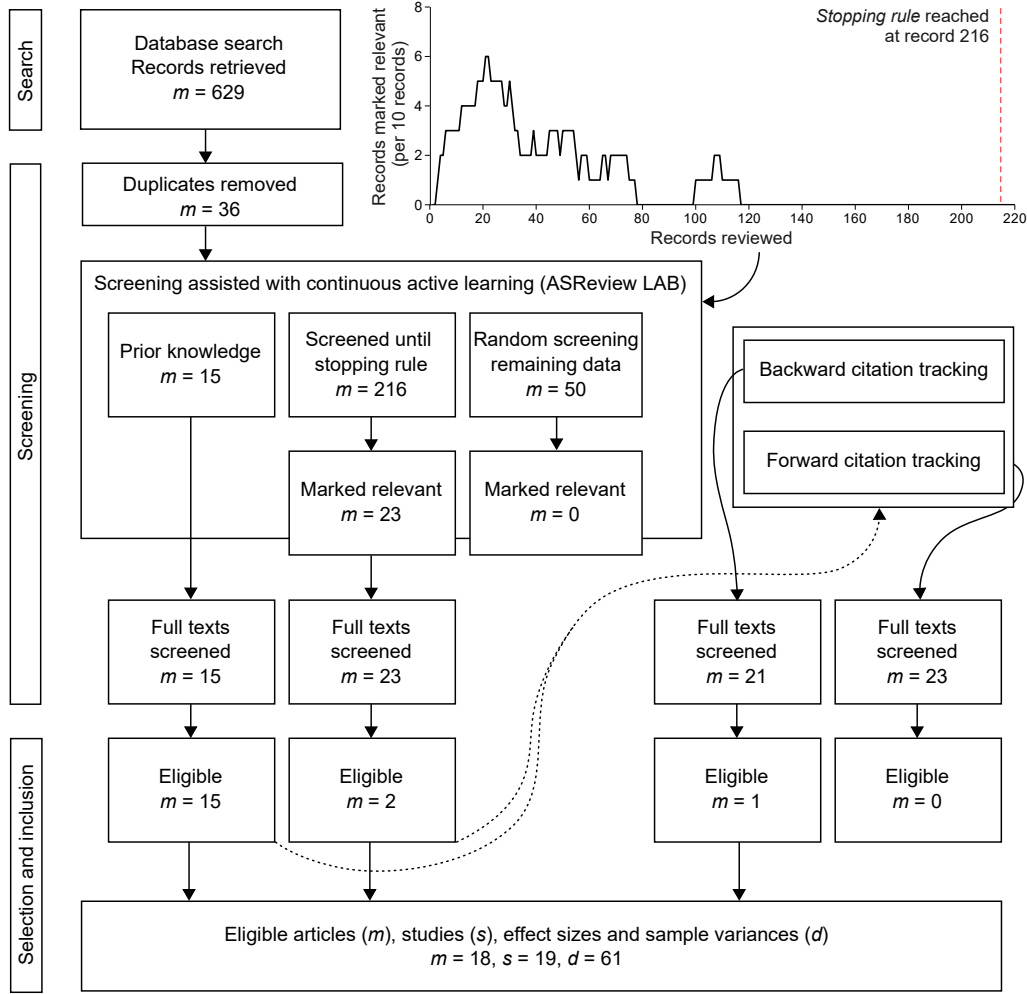


Fig. 1. Overview of the systematic literature review and its results.

variable indicative of homogenization as defined above. Common metrics include pairwise comparisons of cosine distances between transformer-based sentence embeddings based on participant responses, to those of all responses in a sample, e.g., [4, 19, 45], and aggregates of expert ratings of meanings relative to those produced in the sample, e.g., [64]. Surface-level lexical or stylistic similarity metrics, e.g., [32, 53], or where diversity of meanings was captured at the individual-level alone, e.g., [67]), were excluded. Homogenization was measured based on creative output, including divergent thinking, idea generation, writing, visual art and design, music composition, during or after AI use. Peer-reviewed and non-peer-reviewed articles were eligible. Articles had to be available in English, accessible to the authors, and enable computing Cohen's d and sampling variance.

3.3 Screening and Selection

Screening was conducted using ASReview LAB (v2.2), an open-source software for assisting systematic screening through active learning [17]. One author sequentially screened the retrieved records, labeling each presented title and abstract as potentially relevant or not, implementing the eligibility criteria where possible. The active learning model (ELAS u4), using term frequency-inverse document frequency with bigrams, and a linear support vector machine, iteratively predicted and ranked the relevance of unscreened records based on relevant and not relevant records, presenting those with the highest predicted relevance first, and updating the presentation order throughout the screening process. To initialize the model and improve screening accuracy, 15 articles identified as relevant during feasibility scoping and included among the retrieved records were used as prior knowledge. Conservatively, screening continued until 100 consecutive records were labeled non-relevant after the last relevant record [36]. This stopping rule was reached after screening 216 records (Figure 1, top right). In total, 23 additional records were deemed relevant and underwent full-text screening, resulting in two further eligible articles in addition to the 15 previously selected articles. To further minimize the risk of missing relevant articles, 50 records were randomly sampled from the remaining 377 and screened manually. Here, no additional eligible articles were identified.

3.3.1 Backward and Forward Citation Tracking. To cover the search space thoroughly, the 17 eligible articles were used for additional manual backward (screening reference lists) and forward citation tracking (screening citing articles retrieved with Google Scholar’s “Cited by” function). Backward citation tracking yielded 21 articles for full-text screening, of which one met the eligibility criteria. Forward citation tracking yielded another 23 articles for full-text screening, contributing no additional eligible articles. This resulted in a *final sample of 18 articles* included in the meta-analysis. Figure 1 and Table 1 present an overview of the systematic review results and eligible studies.

3.4 Extracting Cohen’s d and Sampling Variance

Cohen’s d and its sampling variance were extracted from reported statistics in most eligible articles. Two exceptions applied: for one study, Cohen’s d and sampling variance were computed directly from the original dataset [48]; for two others [37, 64], they were calculated based on statistics from graphs extracted using Automeris’ WebPlotDigitizer (v5). All effect sizes were (re)coded such that positive values indicated greater homogenization at the collective level in human-AI co-created output. Generally, the somewhat heterogeneous extraction and computation procedures may have led to small deviations from originally reported effect sizes. In total, the 18 eligible articles contributed *19 eligible (quasi-)experimental studies*, and *61 effect sizes* with accompanying sampling variances to be meta-analyzed.

3.5 Data Analysis

A three-level meta-analysis was conducted [62], estimating the pooled effect sizes of homogenization in human-AI co-creation, compared to humans not using AI co-creatively, using R 4.4.1 [49] and metafor 4.6.0 [63]. Since several studies reported multiple homogenization outcomes or for multiple tasks within the same sample, e.g. [70], a three-level approach modeled sampling (Level 1), within-study (Level 2), and between-study variance (Level 3) [62]. Heterogeneity was assessed using the Q -statistic and variance decomposition, indicating unexplained variation in the data. Sensitivity analysis was conducted by re-estimating the three-level model without data from influential studies. Influential studies were detected using study-level Cook’s distances [63], where studies with Cook’s distance above the heuristic threshold $4/k$ were marked as influential (where k is the number of studies). Comparing the results of the complete and sensitivity analyses indicate the extent to which results depend on studies with large effect sizes and sampling variances. Publication

Table 1. Overview of the eligible studies.

Study	Included comparison(s)	Task(s)	Timing	Experiment
[1]	Human vs. human + GPT4o-generated text suggestions	Writing	During	Controlled
[3, 4]	Human + oblique strategies vs. human + access to ChatGPT-3.5	Idea generation	During	Controlled
[5]	Human + 6 human ideas vs. human + 4 GPT-3.5 + 2 human ideas	Divergent thinking	During	Controlled
	Human + 6 human ideas vs. human + 2 GPT-3.5 + 4 human ideas			
[6]	Human vs. human + access to ChatGPT-3.5	Idea generation	During	Controlled
[11]	Human vs. human + GPT-4-generated feedback	Writing	During	Controlled
	Human vs. human + GPT-4-generated example texts			
[19]	Human vs. human + 1 GPT-4-generated example idea	Writing	During	Controlled
	Human vs. human + 5 GPT-4-generated example ideas			
[37]	Human vs. human + GPT4o-generated list of ideas	Divergent thinking	During;	Controlled
	Human vs. human + GPT4o-generated SCAMPER strategies		After	
[38] [†]	Human dyad vs. human + GPT4o prompted to 'collaborate'	Writing	During	Controlled
[42] [†]	Human vs. human + access to GPT-3.5	Writing	During	Controlled
[43] [†]	Human vs. human + access to GPT-4.1	Idea generation	During	Controlled
	Human vs. human + GPT-4.1 ideas + suggestions for elaboration			
	Human vs. human + GPT-4.1 ideas + questions for elaboration			
[45] (2a)	Human vs. human + access to GPT-3.5	Divergent thinking	During	Controlled
[45] (2b)	Human vs. human + access to GPT-3.5	Idea generation	During	Controlled
[46] [†]	Essays written pre-ChatGPT vs. essays written post-ChatGPT	Writing	Unknown	Quasi
[48]	Human vs. human + GPT3 co-writing	Writing	During	Controlled
	Human vs. human + InstructGPT co-writing			
[64]	Human vs. human + Midjourney 4 image idea examples	Visual design	During	Controlled
[65] [†]	Human vs. human + 1x access to ChatGPT o1-preview story plot	Writing	During	Controlled
	Human vs. human + 5x access to ChatGPT o1-preview story plot			
[69]	Visual artworks before vs. after AI adoption	Visual art	Unknown	Quasi
[70] (1)	Articles written pre-ChatGPT vs. articles written post-ChatGPT	Writing	Unknown	Quasi
[70] (2)	Human vs. human + access to ChatGPT-4	Divergent thinking;	During;	Controlled
		Idea generation	After	

Note. Includes study references (study numbers in parentheses), included experimental comparisons, and coded moderators for task, homogenization effect timing, and experiment type. [†] Denotes eligible studies that were not peer-reviewed publications when the systematic review was conducted.

bias was assessed for peer-reviewed studies. Here, funnel plots were generated by plotting study-level Cohen's d , indicating the magnitude of the homogenization affect, against the inverse of the model-implied study-level variance indicating precision, and were visually inspected [63]. An absence of small effects among high-variance studies may indicate selective publication of larger or statistically significant findings. This was complemented statistically [59] by regressing the effect sizes on their standard errors using the three-level approach, where a significant slope may indicate asymmetry. If significant, this warrants further investigation as detailed in [59].

To further understand any homogenization effects, three-level moderator and follow-up meta-analyses were conducted. Moderators were coded by one author across multiple passes through the eligible articles. For SQ1, task types were coded inductively, resulting in four categories: (1) divergent thinking (e.g., alternative uses test), (2) idea generation (e.g., solutions for improving public transport), (3) visual art and design (e.g., character design), and (4) writing (e.g., essay, short story). These categories were entered as levels of a task-type moderator in the three-level model without an intercept. Holm-corrected post-hoc comparisons tested differences between tasks. For SQ2 and SQ3, separate three-level moderator analyses with an intercept were conducted for two binary moderators: (a) whether homogenization effects were measured during human-AI co-creation or after AI use, and (b) whether studies were randomized experiments

conducted in controlled settings or real-world quasi-experiments. If the test of moderation (Q_m -statistic) was not significant, indicating no evidence for differences between moderator levels, a planned follow-up three-level meta-analysis was conducted on the relevant subset of effect sizes (i.e., effects after AI use, or quasi-experimental studies). For these analyses, included effect sizes were restricted to matching task types across moderator levels supporting construct comparability, at the cost of some statistical power. Residual heterogeneity (Q_e -statistic), here, was tested providing insight into whether further unexplained variation in the data persists.

4 Results

4.1 Does Human-AI Co-Creation Lead to Homogenization?

4.1.1 Pooled Effect Sizes. To test whether human-AI co-creation leads to homogenization at the collective level (RQ), a three-level meta-analysis was conducted. The results showed a positive pooled effect size of $d = .334$, 95% CI [.094, .574]. The effect was significant as the confidence interval did not cross zero, and on average small. The test of heterogeneity was significant ($Q(df = 60) = 2850.50, p < .001$), indicating substantial unexplained variation in the data. Variance decomposition suggested that most variation was attributable to between-study variance (approximately 68%) and within-study variance (approximately 31%), and negligible sampling variance. These findings suggest a *homogenization effect* in human-AI co-creation.

4.1.2 Sensitivity Analysis. One influential study contributing three effect sizes was detected within the data. The sensitivity analysis conducted by repeating the three-level meta-analysis without the data from this single influential study showed a slightly smaller positive pooled effect size, $d = .247$, 95% CI [.065, .429]. The effect remained significant as the confidence interval did not cross zero. The test for heterogeneity also remained significant ($Q(df = 57) = 2524.19, p < .001$), still indicating substantial unexplained variation in the data. One percent of the variation was explained at level 1 (sampling variance), 38% at level 2 (within studies), and 60% at level 3 (between studies). The sensitivity analysis therefore strengthens the evidence found in favor of a homogenization effect in human-AI co-creation.

4.1.3 Publication Bias. No evidence of publication bias was found among the peer-reviewed studies (14 studies, 53 effect sizes). Visual inspection of the funnel plot did not suggest any clear asymmetry in study-level effect sizes from peer-reviewed studies relative to their model-implied study-level variances (Figure 2a, dots). One additional observation is that the non-peer-reviewed studies yielded comparably small effects, and, in some cases opposite effects, i.e., stronger homogenization effects without AI use (Figure 2a, triangles). The regression of effect sizes on their standard errors was not statistically significant, $b = 5.36, p = .144$, thus statistically confirming the visual inspection of the funnel plot.

4.2 Do Homogenization Effects in Human-AI Co-Creation Differ Across Creative Task Types?

To test whether homogenization in human-AI co-creation differs across task types (SQ1), a moderator analysis was conducted using a three-level model without an intercept, followed by Holm-corrected pairwise comparisons. The dataset comprised divergent thinking (4 studies, 27 effect sizes), idea generation (5 studies, 14 effect sizes), visual art and design (2 studies, 2 effect sizes), and writing tasks (9 studies, 18 effect sizes).

The test of moderation was significant, $Q_m(df = 3) = 17.12, p = .002$. The homogenization effect was largest for idea generation tasks ($d = 0.70$, 95% CI [.30, 1.10]), representing a medium-to-large effect. Notably, estimates for divergent thinking ($d = 0.12$, 95% CI [−.285, .518]), writing ($d = 0.27$, 95% CI [−.078, .613]), and visual art and design tasks ($d = 0.27$, 95% CI [−.555, 1.088]), however, while also positive, were smaller on average and not statistically significant. The test of

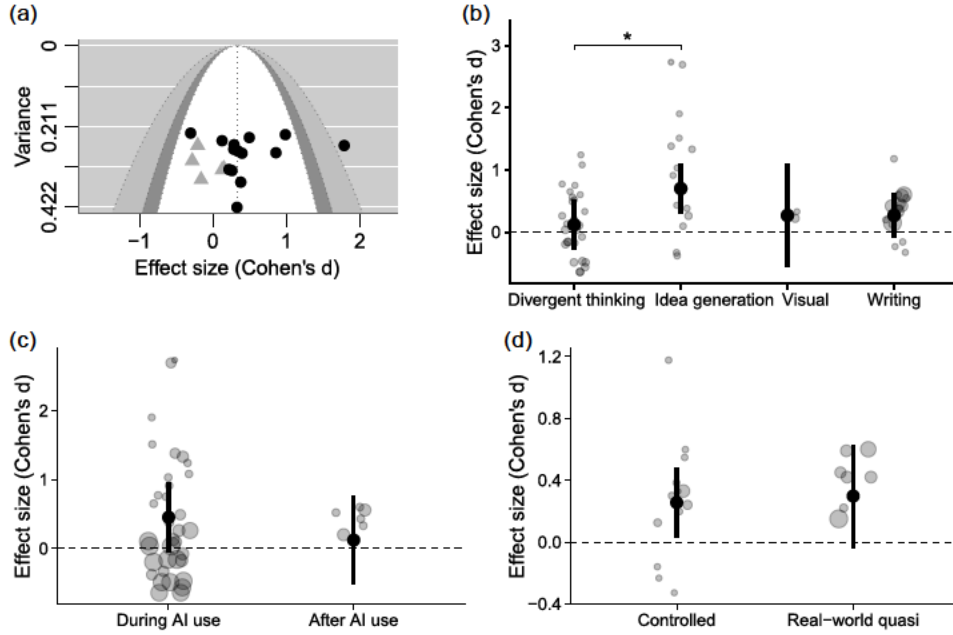


Fig. 2. (a) Funnel plot of study-level effect sizes (Cohen's d) plotted against their model-implied variance. Dots represent peer-reviewed studies, triangles non-peer-reviewed studies, and the vertical line is the pooled effect size. Shaded regions correspond to $p < .10$, $p < .05$, and $p < .01$. (b-d) show dot plots, comparing homogenization effects across tasks (b), effects during and after human-AI co-creation (c), and effects in randomized controlled and real-world quasi-experiments (d). For (b-d), the jittered gray circles are individual effect sizes (Cohen's d), where their size is proportional to the inverse sampling variance of each observed effect, indicating precision. Black dots with vertical error bars represent model-estimated marginal means with 95% confidence intervals. * $p < .05$.

residual heterogeneity was significant, $Q_e(df = 57) = 1860.92$, $p < .001$, suggesting substantial remaining unexplained variation in the data. Holm-corrected pairwise comparisons of the model-implied marginal means indicated that idea generation tasks produced significantly larger homogenization effects than divergent thinking tasks ($M_{diff} = -.582$, $p = .015$). All other pairwise comparisons were not significant (all $p > .050$; see Table 2 and Figure 2b). These findings suggest that homogenization *varies across task types* and appeared strongest for idea generation tasks.

Table 2. Holm-corrected pairwise comparisons of homogenization effects found across the creative tasks

Comparison	M_{diff}	se	Z	p	CI.LB	CI.UB
Divergent thinking – Idea generation	-0.582	0.193	-3.02	0.015	-0.960	-0.204
Divergent thinking – Visual art and design	-0.150	0.466	-0.32	1.000	-1.064	0.764
Divergent thinking – Writing	-0.151	0.270	-0.59	1.000	-0.680	0.379
Idea generation – Visual art and design	0.432	0.466	0.93	1.000	-0.481	1.345
Idea generation – Writing	0.432	0.269	1.61	0.542	-0.095	0.958
Visual art and design – Writing	-0.001	0.455	-0.00	1.000	-0.892	0.891

Note. Data are the differences between the model-implied marginal means (M_{diff}), standard errors (se), z-values (Z), p-values (p), and the lower (CI.LB) and upper (CI.UB) bounds of the 95% confidence intervals. Positive values indicate larger homogenization effects for the first task type listed in each contrast.

4.3 Do Homogenization Effects Persist After Episodes of Human–AI Co-Creation End?

To test whether homogenization effects persist after episodes of human–AI co-creation end, indicating effects that extend beyond immediate co-creative engagement (SQ2), the dataset was curated to include effect sizes that could be coded as reflecting homogenization either during co-creation or after obtained from studies administering divergent thinking and idea generation tasks, since post human–AI co-creation effects in the dataset were only measured using these two types of tasks. The resulting dataset comprised eight studies (35 effect sizes) assessing homogenization effects during human–AI co-creation and two studies (6 effect sizes) assessing effects after AI use. The latter two studies contributed effect sizes to both moderator conditions [37, 70].

The test of moderation was not significant, $Q_m(1) = 2.17, p = .141$. The intercept reflects the pooled effect specifically during co-creation, $d = .453$, 95% CI [.048, .953], while the contrast between during and after conditions was negative but non-significant, $b = -.331$, 95% CI [−.771, .110] (Figure 2c). The test of residual heterogeneity was significant, $Q_e(df = 39) = 1111.59, p < .001$, indicating substantial remaining unexplained variation in the data. This finding offers no evidence for a difference in homogenization during and after human–AI co-creation. The three-level meta-analysis on the data for homogenization effects after human–AI co-creation alone showed a positive pooled effect size, $d = .414$, 95% CI [.235, .593]. These results showed a small-to-medium effect size for post AI-use effects that was significant. The test for heterogeneity was not significant $Q(df = 5) = 5.75, p = .331$. Variance decomposition suggested that most variation was attributable to sampling variance (approximately 70%) and within-study variance (approximately 30%), with negligible between-study variance. This finding, though tentative due to the small number of studies included, suggests that homogenization effects *might persist* beyond immediate human–AI co-creative engagement.

4.4 Are Homogenization Effects Limited to the Lab, or Do They Extend to Real-World Creative Contexts?

To test whether homogenization in human–AI co-creation is primarily a lab phenomenon or also extends to real-world creative contexts (SQ3), the dataset was curated to include controlled experiments and real-world quasi-experiments. Because the quasi-experiments contained data only from writing and visual tasks, data from other task types were excluded to support construct comparability. The resulting dataset comprised eight studies (13 effect sizes) from controlled experiments and three studies (7 effect sizes) from quasi-experiments.

The test of moderation was not significant, $Q_m(df = 1) = .05, p = .837$. The intercept reflects the pooled effect in controlled experiments, $d = .255$, 95% CI [.034, .477], while the contrast with quasi-experiments was not significant, $b = .042$, 95% CI [−.358, .442] (Figure 2d). The test of residual heterogeneity was significant, $Q_e(df = 18) = 1017.61, p < .001$. This finding offers no evidence for a difference between controlled and quasi-experiments. The three-level meta-analysis on data for real-world quasi-experiments alone, however, showed a positive pooled effect size, $d = .303$, 95% CI [.082, .524]. The effect was significant as the confidence interval did not cross zero and represented a small effect. The test of heterogeneity was significant $Q(df = 6) = 938.71, p < .001$. Variance decomposition suggested that most variation was attributable to between-study variance (approximately 80%) and within-study variance (approximately 20%), with negligible sampling variance. This finding, though tentative due to the small number of studies, suggests homogenization effects in human–AI co-creation *may extend* to real-world contexts.

5 Discussion

This meta-analysis examined whether human–AI co-creation leads to homogenization of creative outputs at the collective level (main RQ), and how such effects vary across creative task types, time, and context (SQ1–SQ3).

5.1 Does Generative AI Make Us Think Alike?

The findings indicate that engaging in co-creative interaction with GenAI tends to reduce diversity of meanings across creative outputs. This pattern aligns with a growing body of empirical work showing a recurrent trade-off in AI-assisted creativity: creative gains at the individual level are often accompanied by convergence of meanings at the collective level, e.g., [19, 39, 45]). In this perspective, homogenization does not signal a failure of creativity per se, but rather a *reconfiguration* of *where* novelty appears and *how* it is distributed across participants.

The size of the pooled effect and associated heterogeneity suggest that homogenization should not be interpreted as an immediate collapse into uniformity. Rather, it reflects a gradual *distributional shift* emerging through repeated interactions with similar GenAI systems. Creative outputs may remain novel or valuable in isolation, even as the collective breadth of ideas, styles, or meanings narrows. From a systemic and sociocultural creativity perspective, this pattern is consequential: creative systems can remain locally productive while becoming globally less diverse over time, thereby constraining future opportunities for creative exploration and transformation [10, 25, 31].

These findings highlight that the key research question is not simply whether human-AI co-creation leads to homogenization, but more precisely under which *conditions*, through which, and with what temporal and contextual *boundaries* such effects arise. Rephrasing the research question shifts the analytical focus from the presence of homogenization to how it *unfolds* across tasks, over time, and beyond controlled lab settings.

5.2 Task Sensitivity, Temporal Persistence, and Ecological Validity (SQ1–SQ3)

5.2.1 Task Sensitivity (SQ1). The results show that homogenization effects differ across creative task types. Homogenization was most pronounced in ideation tasks (e.g., proposing solutions to improve public transport [4]), where semantic framing and evaluative constraints are relatively strong, and significantly exceeded effects observed in divergent thinking tasks (e.g., generating alternative uses for a paperclip [37]). This pattern is consistent with creativity research suggesting that tasks involving early constraint and evaluation are more susceptible to convergence of meanings than open-ended tasks emphasizing associative exploration, e.g., [7, 31]. From an HCI perspective, ideation tasks are also typically characterized by iterative shifts between exploration and evaluation, often occurring within tightly framed problem spaces [34]. When such processes are mediated through predominantly *linear* AI interaction workflows, sequential path dependence may further amplify convergence by making earlier framings and model suggestions disproportionately salient [68]. This helps explain why homogenization effects appear more pronounced in AI-assisted ideation than in minimally constrained divergent thinking tasks. Differences among the remaining task comparisons were not statistically significant and so provided no strong evidence for additional task-specific distinctions.

5.2.2 Temporal Persistence (SQ2). The findings further suggest that homogenization effects may persist *beyond immediate episodes* of human-AI co-creation. While no statistically significant difference was observed between effects measured during and after AI use, the pooled effect for post-AI conditions alone was small-to-medium and statistically significant. It should be noted that in the studies assessing post-AI use effects, homogenization was often measured in subsequent instantiations of similar tasks, rather than as a simple continuation of the same creative episode [37, 70]. This suggests that what carries over may be a *narrowing in creative thinking* or *semantic framing*, rather than persistence within a single ideation process. Such findings are consistent with cognitive accounts of fixation and example-based interference [9, 27, 57]. This evidence remains tentative due to the limited number of studies available.

5.2.3 Ecological Validity (SQ3). Finally, the results indicate that homogenization effects are not confined to lab settings. Although no moderation by experiment type was detected, analysis of quasi-experimental studies alone yielded a small but significant pooled effect, indicating homogenization in writing and visual art and design tasks in real-world contexts. This makes it plausible that homogenization may also manifest in everyday creative practices [46, 69, 70]. However, conclusions regarding *ecological validity* remain provisional given the small number of eligible studies. Across all three SQs, significant (residual) *heterogeneity* remained, indicating that important factors, e.g., prompting conventions, interaction design, collaboration structure, or model diversity, are not yet well-understood. This underscores the need for further empirical work to examine when and how homogenization emerges in human–AI co-creation.

5.3 Implications for HCI and Creative Practice

The findings suggest that homogenization in human–AI co-creation is best understood as a *sociotechnical, task-dependent, and temporally extended phenomenon*. Here, homogenization emerges neither solely from the properties of GenAI nor solely from user behavior, but from the *repeated coupling* of human creative processes with pre-trained GenAI operating on shared data distributions and optimization objectives, cf. [50]. Recent work on *creative control* in designer–AI co-creation further suggests that how agency and decision authority are allocated within these systems can simultaneously enhance individual creative performance while constraining exploratory breadth [26]. Homogenization may arise not despite effective human control, but partly because well-tuned AI assistance supports efficient convergence of meanings under conditions of shared framing. In this sense, homogenization may be better understood as an *emergent property* of human–AI co-creative systems rather than as a failure caused by any single component.

The homogenization effects observed in human–AI co-creation do not indicate that humans are becoming less creative in any absolute sense. Rather, they suggest that when creative activity is increasingly mediated by GenAI systems optimized for fluency and typicality, *creative diversity* at the *collective* level may erode even as local performance improves. As Lockhart [40] has recently argued, “What is often celebrated as AI creativity is better described as functional performance under conditions of maximal constraint: creativity without ownership, expression without interiority, novelty without responsibility” (p.1). This implies that homogenization is not a paradoxical side effect of GenAI, but a *predictable outcome* of generative systems designed to optimize output without human-like agency.

These implications extend beyond individual interactions. As GenAI tools become embedded across education, research, and creative industries, even modest distributional shifts may accumulate into broader patterns of cultural convergence or monoculture [47, 61]. GenAI thus functions not only as a creativity support tool (CST) in a traditional HCI sense, e.g., [21, 22, 54, 55], but as a *cultural technology* reshaping how novelty, authorship, and value are produced and evaluated at scale [35, 50]. Designing *against homogenization* requires attention not only to user-interface level interventions, but to the sociotechnical infrastructures and evaluative regimes in which human–AI co-creation unfolds.

5.4 Limitations

Several limitations should be considered when interpreting these findings. First, the meta-analysis primarily reflects homogenization effects associated with *pre-trained GenAI systems*, particularly LLMs. The findings therefore may not generalize to other forms of human–AI co-creation, such as adaptive machine-learning (ML) systems, embodied AI, or long-term collaborative engagements [18, 50]). Second, the *range of creative domains* examined remains limited. Most eligible studies investigated divergent thinking, ideation, writing, and visual art or design, with no studies on music or audio composition meeting the inclusion criteria, cf. [33]. This constrains the generalizability of the results. Third, several analyses, particularly those relevant to SQ2 and SQ3, were based on *relatively small subsets of studies*. Therefore,

conclusions regarding temporal persistence and ecological validity should be regarded as provisional. Finally, many studies relied on linear, dialogue-based GenAI interaction, which may bias creative processes by reinforcing sequential path dependence and limiting parallel exploration [68]. How alternative interaction structures, such as nonlinear or canvas based workflows, might mitigate or reshape homogenization therefore remains an open question.

6 Conclusion and Future Work

This meta-analysis provides the first systematic synthesis of empirical evidence on homogenization effects in human-AI co-creation. The findings offer qualified answers to the main research question (RQ) and its three complementary sub-questions (SQ1–SQ3), clarifying when, how, and under which conditions human-AI co-creation may reduce creative diversity at the collective level. Instead of framing homogenization as an inevitable consequence of GenAI, the results position it as a *conditional and context-dependent phenomenon*, shaped by task structure, temporal dynamics, and sociotechnical feedback processes, e.g., [29, 44]. Rather than signaling a loss of creativity, the findings point to a *gradual reorganization of creative diversity* in AI-mediated systems: subtle in isolated instances, but consequential when examined at scale and over time. Across the reviewed literature, homogenization appears most pronounced in AI-assisted ideation characterized by structured problem framing, linear interaction, and optimized creative control, suggesting a conditional, design-dependent outcome rather than an inherent property of generative models [26, 34, 68].

For the ECCE and broader HCI research community, these findings underscore the importance of studying creativity not only at the level of individual interaction, but as a property of human-AI co-creative systems operating over extended periods, especially in real-world contexts. Future work should combine controlled lab experimentation with longitudinal, large-scale field studies to better understand how creative trajectories evolve in AI-mediated contexts, e.g., [5, 53]. In our view, this is the premise for developing creativity support systems that sustain diversity at the collective level, and it will require advances not only in interaction design, but also in sociotechnical governance and model design. By examining whether GenAI may make us think alike, this paper aims to offer a step toward this goal.

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